

01 Introduction

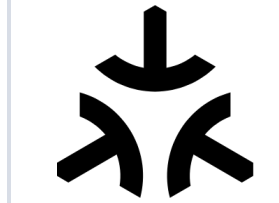
Smart home technology has expanded rapidly, but many modern systems remain expensive, difficult to configure, and dependent on cloud services or dedicated infrastructure. Platforms such as Home Assistant and Homebridge offer advanced customization but require additional hardware, networking knowledge, and constant software upkeep.

This project investigates whether a compact, modular “smart puck” can offer a more accessible alternative using Matter over Thread, connecting relays, buttons, LEDs, and sensors to modern ecosystems while remaining low-cost, low-power, and customizable.

02 Research Goals

- Evaluate whether modern embedded microcontrollers can reliably support Matter over Thread.
- Test long-term device stability and persistent operation after reboot.
- Verify compatibility with Zephyr RTOS for low-level hardware control.
- Support modular GPIO expansion for relays, buttons, sensors, and other peripherals.
- Identify limitations and challenges within the current Matter and Thread ecosystems.
- Analyze the balance between connectivity, reliability, and real-world deployment practicality.

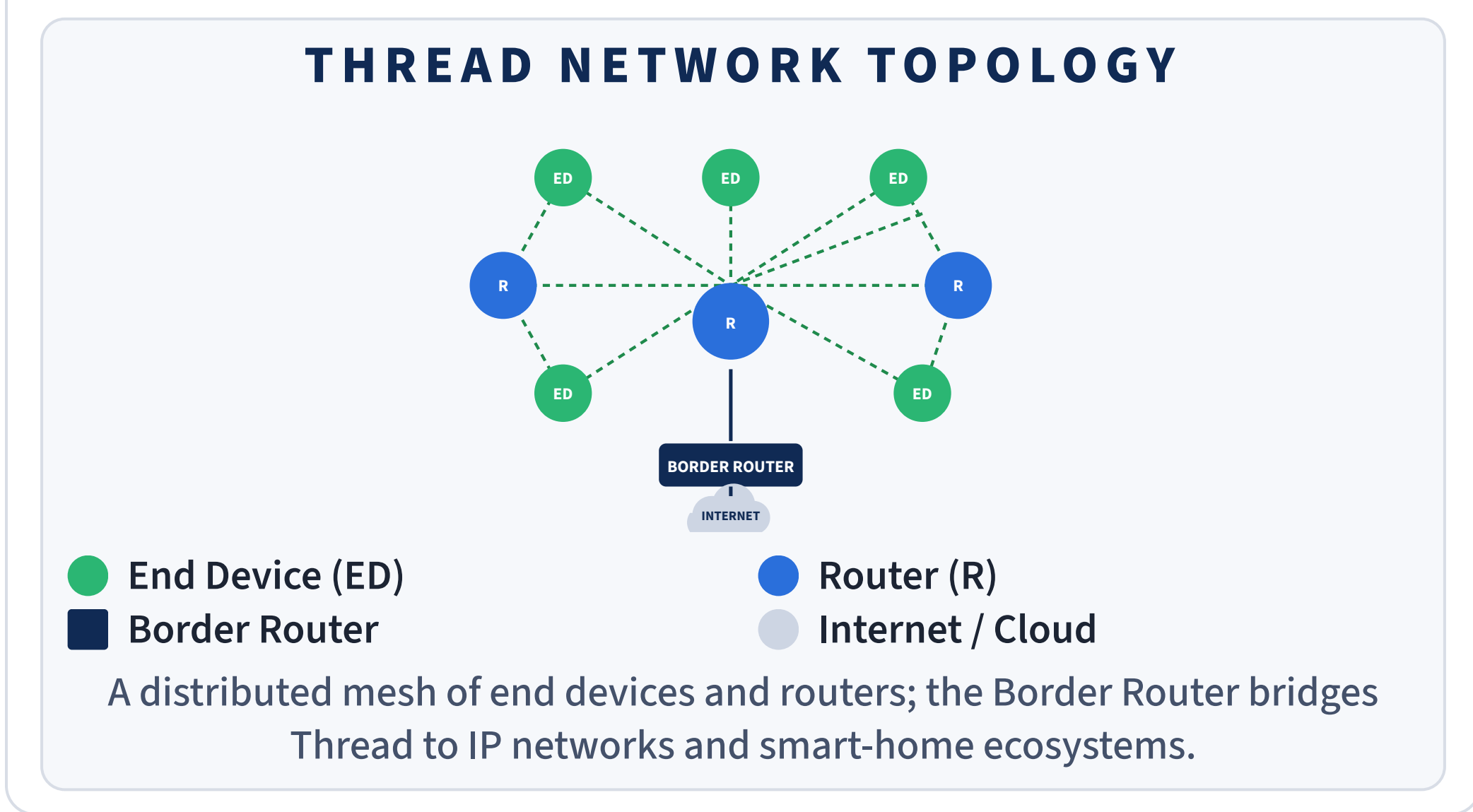
03 Background

 **Matter**

An open smart-home standard designed to improve interoperability across Apple Home, Google Home, Alexa, and SmartThings. Matter uses standardized device “clusters” and secure local communication, with simple commissioning via QR code or BLE.

 **Thread**

A low-power IPv6 mesh protocol built on IEEE 802.15.4. Thread offers self-healing routing, low energy use, and scalability, but requires a Thread Border Router to bridge the mesh to the wider IP network.



SMART PUCK

Evaluating Matter over Thread for Modular Smart Home Devices

PROJECT GOAL

Develop a compact, low-cost smart puck capable of integrating simple devices into Matter-enabled smart-home ecosystems while remaining modular, reliable, and easy to configure.

04 Methods

Two embedded hardware platforms, the Arduino Nano Matter and the Seeed Studio XIAO nRF54L15, were evaluated under Zephyr RTOS with Matter over Thread workflows. Firmware was built with west, flashed to each target, and monitored via serial logging. Testing emphasized long-term runtime behavior, not just initial commissioning success.

Platforms & Tooling

- Zephyr RTOS with the Nordic Connect SDK as the base development environment.
- chip-tool used for Matter commissioning and cluster testing.
- Apple Home and Google Home as production smart-home controllers.
- Raspberry Pi running OpenThread as the Thread Border Router.

EVALUATION METRICS

- Commissioning success
- Reboot persistence
- Runtime stability
- Driver flexibility
- GPIO support
- Development practicality

05 Results & Key Findings

Arduino Nano Matter

Vendor-supplied Matter examples compiled, flashed, and commissioned successfully. However, the platform failed under Zephyr RTOS, which is required for the modular architecture, due to missing IEEE 802.15.4 driver support, incomplete device-tree configuration, and restricted HAL access that blocked custom GPIO and relay drivers.

XIAO nRF54L15

The XIAO nRF54L15 paired cleanly with Zephyr and the Nordic Matter SDK. Matter commissioning, Thread attachment, and custom GPIO functionality all worked, with strong low-level hardware access for modular embedded development.

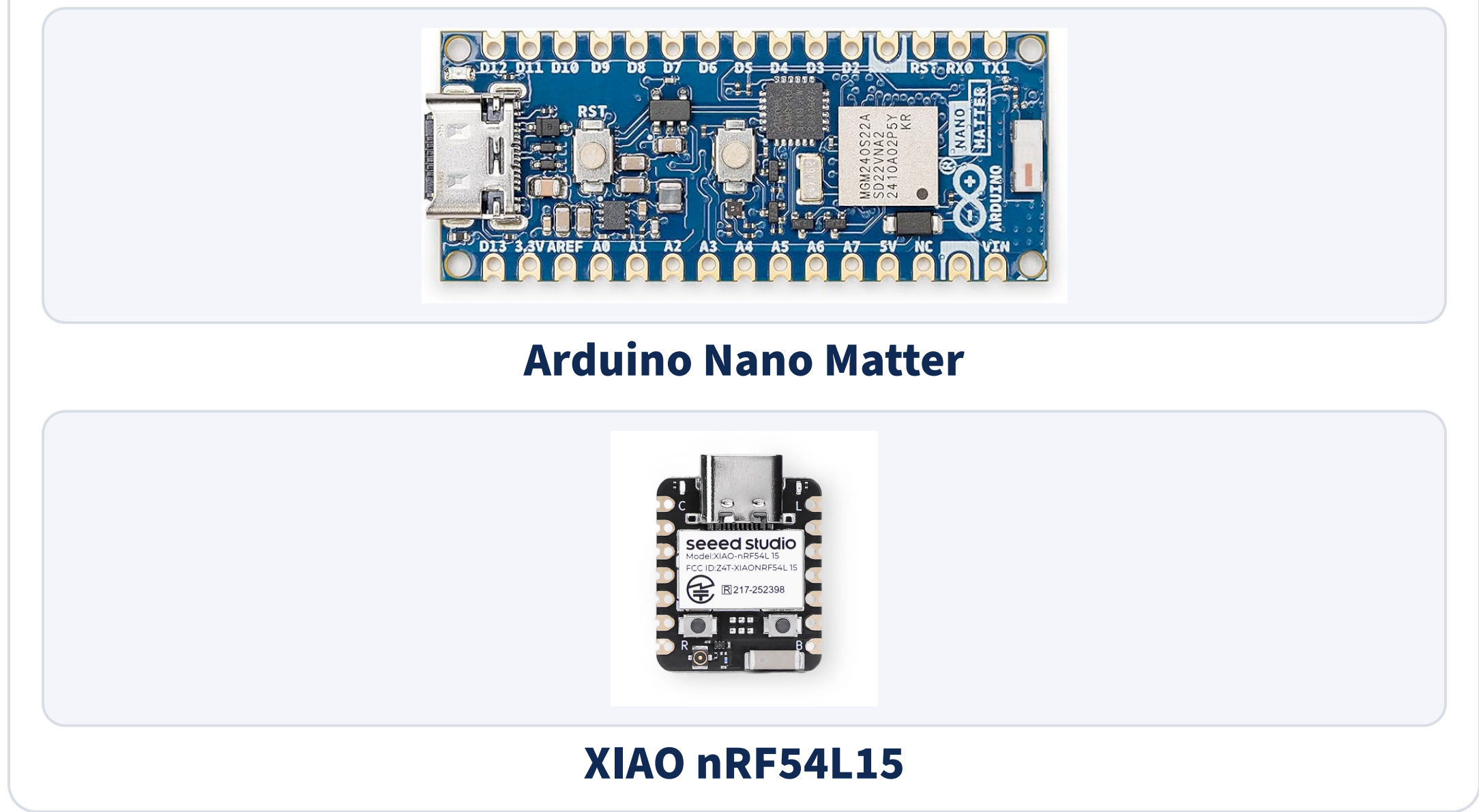
Runtime Stability & Outcome

Reboot testing exposed repeatable crashes ~40 to 47 seconds after boot, traced to MPSSL ASSERT faults in radio scheduling. Logs confirmed credentials and Thread attach succeeded before each crash. Tuning HFCLK latency and adding custom assert handlers improved survivability but did not eliminate the fault.

Despite the stability ceiling, the XIAO nRF54L15 offered the strongest balance of functionality, flexibility, and development support, and was selected as the final hardware platform for continued Smart Puck development.

06 Hardware

Two integrated 802.15.4 development boards, compared on SDK maturity, Zephyr support, driver access, and runtime stability.



07 Final Build Error

```
<err> os: ***** HARD FAULT *****
<err> os:   Fault escalation (see below)
<err> os: ARCH_EXCEPT with reason 3

<err> os: r0/a1:  0x00000003  r1/a2:  0x20023fa7  r2/a3:  0x00000000
<err> os: r3/a4:  0x00000003  r12/ip: 0x00000000  r14/lr:  0x0002289f
<err> os: xpsr:  0x010000f5
<err> os: Faulting instruction address (r15/pc): 0x000333be
<err> os: >>> ZEPHYR FATAL ERROR 3: Kernel oops on CPU 0
<err> os: Fault during interrupt handling

<err> os: Current thread: 0x20013020 (idle)
<err> os: Halting system
0015910 C Error: [Errno 32] Pipe error [ _main ]
```

HARD FAULT · ZEPHYR FATAL ERROR 3, fault escalation during interrupt handling on the XIAO nRF54L15. Repeated **MPSSL ASSERT** failures caused crashes 40 to 47 s after restart.

08 Conclusions

Matter over Thread is technically achievable on modern embedded platforms, but current implementations still face significant limits in reliability, portability, and development flexibility.

Vendor-supported environments simplify setup but restrict low-level access. RTOS-based development through Zephyr exposes greater hardware control at the cost of driver maturity, timing constraints, and runtime stability.

Successful commissioning alone is not enough: reboot persistence, fault recovery, and stable runtime behavior proved equally critical when assessing platform viability.

The Seeed Studio XIAO nRF54L15 ultimately provided the strongest foundation for continued development.

09 Future Work

- Continued work will focus on long-term runtime stability and fault recovery on the XIAO nRF54L15, with deeper debugging of MPSSL ASSERT failures, radio-scheduling optimization, and watchdog-based recovery.
- Expand modular GPIO and peripheral support.
 - Improve low-power and deep-sleep operation.
 - Finalize enclosure and hardware layout.
 - Develop a simplified mobile setup workflow.
 - Test with real smart-home automations.
 - Implement OTA firmware update support.

10 Key Takeaways

- Matter compatibility ≠ development flexibility.
- Reboot persistence is critical for reliable IoT deployment.
- Driver maturity strongly impacts embedded viability.
- Real-world runtime testing reveals issues hidden in demos.
- Embedded development means balancing stability, cost, flexibility.
- Ecosystem maturity and documentation shape development speed.

11 References

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